

Convolutional Neural Networks (CNN)

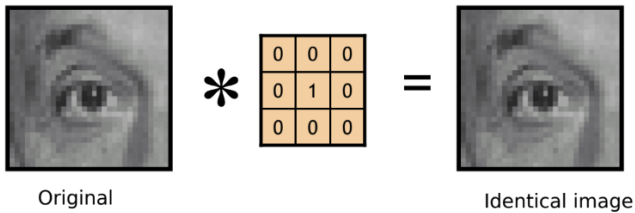
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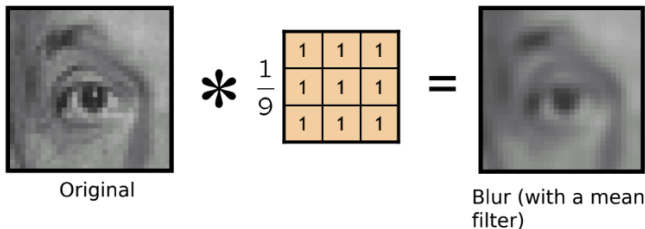
Purpose of Convolution

Convolution extracts local spatial features from the input image using small filters. Each filter detects specific patterns (edges, textures, corners), forming hierarchical feature representations.

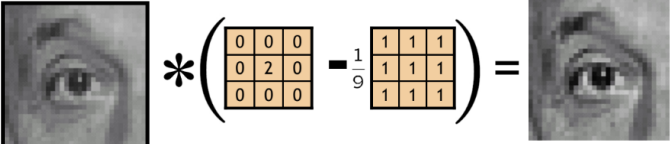
Convolution Filters(Identity Filter)



Convolution Filters(Average/Blur Filter)



Convolution Filters(Sharpen Filter)

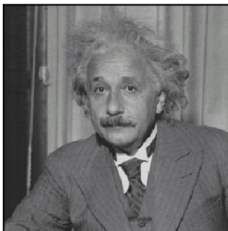


The diagram illustrates the sharpening filter operation. It shows an original grayscale image of an eye being convolved with a sharpening filter (Laplacian filter) to produce a sharpened image. The filter is represented as a 3x3 matrix with a central value of 2 and surrounding values of -1, scaled by 1/9.

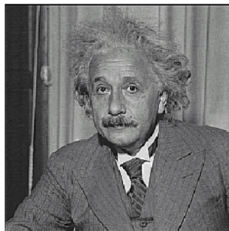
$$\text{Original} * \left(\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right) = \text{Sharpening filter (Laplacian filter)}$$

Original

Sharpening filter (Laplacian filter)

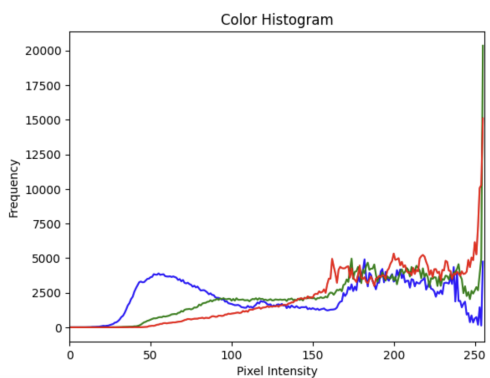


before



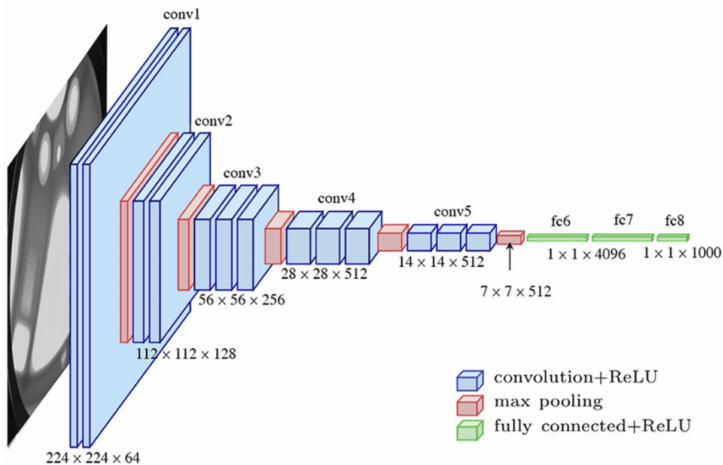
after

Color Histogram



The color histogram displays the three primary color channels present in our image. The x-axis (horizontal) represents the pixel intensity values ranging from 0 to 255 for each color channel.

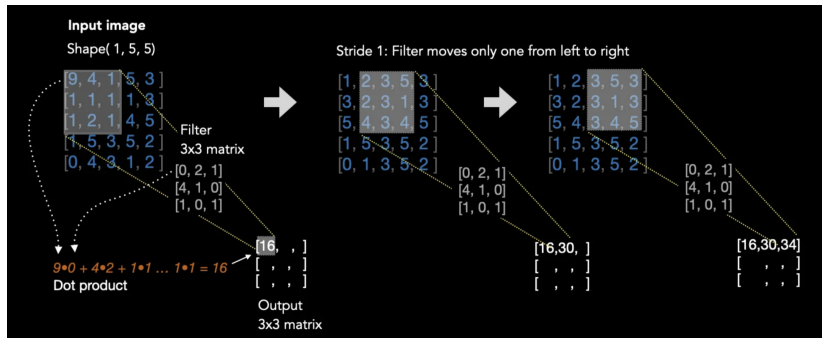
CNN



Why CNNs for Images?

- Exploit spatial locality.
- Fewer parameters than fully connected networks.
- Weight sharing makes models more efficient.

CNN filters



Necessity of Multiple Filters

- Capture diverse features such as edges, colors, and textures.

Padding and Stride

- **Padding:** Adds zeros around borders to preserve dimensions (*same padding*).
- **Stride:** Defines the step size of the filter. Larger stride reduces output resolution.

- Reduces spatial dimensions and introduces translation invariance.

Example: Max Pooling and Small Shift

Before shift:

$$\begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$$

Max Pooling (2x2) $\rightarrow$ 3

After small shift:

$$\begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$$

Max Pooling (2x2) $\rightarrow$ 3

Observation: Output remains the same despite small translation.
This makes CNNs robust to feature position changes.

Max vs Average Pooling

- **Max Pooling:** Takes strongest activation.
- **Average Pooling:** Takes mean activation.
- Max pooling preserves salient features; average pooling smooths output.

- **ReLU:** Introduces non-linearity, avoids vanishing gradients.
- **Multiple Filters:** Capture diverse features such as edges, colors, and textures.

- **Translation Invariance:** CNNs recognize features regardless of position.
- **Weight Sharing:** Same filter weights applied across all spatial locations.